

Exploring Human-Computer Interaction (HCI) criteria in the design and assessment of Next Generation VR based education and training environments

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ABSTRACT

This paper discusses the approach and outcomes of adopting design and assessment criteria based on Human-Computer Interaction (HCI) principles. A general framework is presented which has been adapted to support teaching and training in two domains: first to teach science and engineering concepts to students with autism and second to support the training of first responders involved in the COVID-19 pandemic. The proposed framework emphasizes the importance of HCI principles such as affordance, visual density, and cognitive load during the design process. The training environments were created using various interfaces and immersion levels; these included haptic based interfaces and fully immersive 3D environments. Assessment activities were conducted to gain a better understanding of the impact of such a cyber-human framework. The preliminary results of the assessment activities demonstrated the positive impact of such cyberlearning techniques and environments for both the domains.

Keywords: Human-Computer Interaction (HCI), VR, COVID-19, Autism.

Index Terms: Human-centered computing—Human-computer interaction (HCI)—Interaction paradigms—Virtual Reality

1 INTRODUCTION

Virtual Reality (AR) can be considered as one of the emerging technologies that can support learning in engineering [1], medicine [2], and education [3]. This paper presents the design and development of a cyber-human framework based on Human-Computer Interaction (HCI) principles for two domains. The first domain explores the use of VR in teaching science and engineering concepts to students with Autism and the second domain focuses on the training of first responders involved in the COVID-19 pandemic.

HCI is an attempt to address the pressing industrial and societal need for advanced education and research in humanizing computer-based systems. The focus of this paper is to investigate the role of affordance, visual density, and cognitive load during the design process of the VR environments and subsequently, their effect on users' skills and knowledge acquisition after interaction with the VR environments. In the context of Human-Computer Interaction, the term affordance was defined by Norman as action possibilities

that are perceivable readily by an actor [4]. We proposed dynamic affordance (DA) which can be defined as a function of comprehension of a scene by a user inside a virtual 3D environment moving along a specific path P (within that target 3D environment) over a fixed period of time (T). Further, we propose visual density to refer to the number of objects/cubic unit in the environment. The objective is to study the role of visual density on the affordance of a virtual environment. Cognitive load refers to the working memory load utilized by a user when performing a particular task [5]. In learning complex tasks such as flying airplanes and performing surgeries, the cognitive load becomes a crucial factor.

COVID-19 has placed an overwhelming strain on our Nation's ability to treat patients. Currently, there is an urgent need to train such responders (nurses, physician assistants) to perform the testing and ventilator activities in a safe and efficient manner. The cyber human framework is adapted to design a VR training simulator with two objectives: (i) increase the pool of first responders involved in COVID-19 testing and using ventilators on patients (ii) develop a more effective (and less risky) process to train and prepare such first responders.

Autism can be described as a developmental disability that can cause significant social, communication, and behavioral challenges [6]. In today's contexts, there is a need to explore the potential of cyber-human frameworks based on VR and haptic technologies to support science learning for children with autism. The cyber-human framework is also adapted to teach the concepts of science and engineering to students with autism.

2 DESIGNING THE CYBER-HUMAN FRAMEWORK BASED ON HCI PRINCIPLES AND PARTICIPATORY DESIGN APPROACHES

The participatory design approach [7] was used in the design of our cyber-human framework for VR based training and education. This is useful because the experts and professionals in the field have a greater understanding of standard and non-standard working environments, and as such would understand which of those environments would be the most familiar and efficient in training.

3 THE CREATION OF THE VIRTUAL LEARNING ENVIRONMENTS FOR SCIENCE EDUCATION

Several VLE based modules were developed (built) for autistic students to teach concepts in assembly, robotics, density, and manufacturing. In this paper, the focus of interest is the design of additional VLEs with the scope of learning extended to space systems (introduction to NASA's Moon Mission and related concepts in path planning and assembly). Medical and educational specialists agree that children with autism and teens learn best using Applied Behavior Analysis (ABA) [8], which follows a single subject experimental analysis of behavior design. Positive reinforcement provides opportunities for rewards when participants perform desired learning behaviors. In the NASA Moon Mission

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modules, the students were introduced to the concepts of path planning and assembly using NASA's Deep Space Mission contexts (pertaining to the Artemis 2024 mission to land on the Moon) (Fig. 2). During all learning interactions, positive learning behavior was emphasized through congratulatory messages and rewarded through positive reinforcers. VLEs have been created and assessed for middle and elementary school students (both girls and boys participants) [3, 9, 10].



Figure 1: A student interacting with the VLE using the Vive immersive headset and controller and Figure 2: View of a learning scenario inside the VLE where a teaching avatar (left) explains concepts of disassembly and service using a laptop (inside the Gateway structure)

Assessment of affordance

TEST 1: 2 groups of 4 students (2 middle and 2 high school students) were involved. One group interacted with scenes with lower visual density; the other with scenes of high visual density. Affordance was measured based on the understanding of scene tasks and target concepts learned through interactions.

Null Hypothesis: Visual Density does not influence Affordance and Learning. Based on the t-test, the null hypothesis was rejected.

TEST 2: Introduce interruptions and disturbances to study the impact on student learning; ambulance siren, someone talking in the back of the room (in the scene); 2 groups of three middle school students were involved: one group interacted without interruptions and distractions; the second group interacted with interruption and distractions. The assessment focused on how such interruptions and distractions affected understanding and grasp of the target learning concepts for autistic students.

Null Hypothesis: Interruptions and distractions do not influence Affordance and Learning. Based on the t-test, the null hypothesis was rejected.

Finding #1: students who interacted with low visual density environments demonstrated high affordance, measured by the understanding of target concepts compared to those who did not receive any reinforcers

Finding #2: students who interacted with VLEs without disturbances scored higher on the post-tests of knowledge regarding assembly and path planning

Results of Positive Reinforcers

Null Hypothesis: Positive reinforcers do not influence learning. Based on the t-test, the null hypothesis was rejected.

Finding: The improvement in the group who received positive reinforcers is higher compared to the group who did not receive any reinforcer.

4 THE CREATION OF THE VR BASED TRAINING ENVIRONMENTS FOR FIRST RESPONDERS

The VR based training environments for first responders were developed for the Vive fully immersive platform. Three training environments developed for the first responders training focused on the COVID-19 pandemic to train first responders in pre-swab, swab, and post-swab activities.

The assessment activities were conducted at a medical center. The expert guided us in the design and development of the VR environments helped conduct the assessment activities.

Assessment Activities

Twelve participants were divided into two groups. The first group (group A) interacted with the training environments without

additional cognitive load and the second group (group B) interacted with the training environments with the additional load.

Null Hypothesis: Cognitive load does not influence knowledge acquisition. Based on the t-test, the null hypothesis was rejected.

Finding: The group without cognitive load (Group A) showed higher overall improvement compared to the group with additional cognitive load (Group B).

5 CONCLUSION

This paper discussed an innovative VR based cyber-human framework that was developed following the HCI principles. The generalized framework was adapted for two domains; the first domain was teaching science concepts to students with autism and the second domain was the first responder training for the COVID-19 pandemic. A discussion of HCI principles such as affordance, visual density, and cognitive load which was used during the design of the cyber-human framework was also presented in the paper. Such a framework was developed using the participatory design approach which was also elaborated. Assessment activities were conducted and the results which underscore the importance of using VR-based training environments for training and teaching were also presented in the paper.

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